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FORM TP 2013005

JANUARY 2013

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE® EXAMINATION

CHEMISTRY

Paper 02 – General Proficiency

2 hours and 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX compulsory questions in TWO sections.
2. Write your answer to EACH question in the space provided in this answer booklet.
3. Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.
4. Return this booklet at the end of the examination.
5. You may use a silent, non-programmable calculator to answer items.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

Answer ALL questions.

Write your responses in the spaces provided in this answer booklet.

Do NOT spend more than 30 minutes on Question 1.

1. Derrick performed four experiments to investigate how certain factors affect the rate of production of hydrogen gas when zinc metal reacts with dilute hydrochloric acid. In Experiment 1, which was done at 30 °C, he added 1.0 gram of granulated zinc in excess to 0.1 mol dm^{-3} hydrochloric acid and measured the volume of hydrogen gas liberated after 2 minutes. He then performed three other experiments with the following variations: In Experiment 2, he varied the concentration of the acid; in Experiment 3, he varied the form of zinc, and in Experiment 4, the temperature. In EACH experiment (illustrated in Figure 1), the gas produced was collected for the **same** period, 2 minutes. The data for Experiments 1, 2, 3 and 4 are summarized in Table 1.

TABLE 1: DATA FOR EXPERIMENTS

Experiment	[HCl] (mol dm^{-3})	Form of Zinc	Temperature (°C)	Volume (cm^3)
1	0.1	granules	30	89
2	0.2	granules	30	
3	0.1	powder	30	
4	0.1	granules	20	

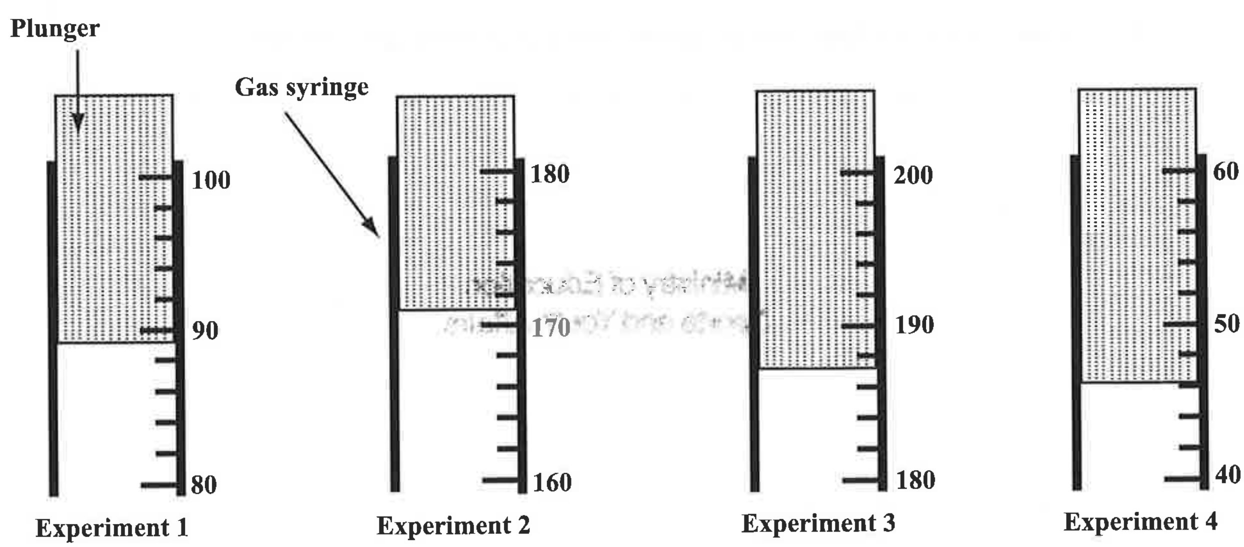

Figure 1. Diagrams showing the volumes, in cm^3, of gas produced as seen on a gas syringe

(a) Define the term 'rate of reaction'.

____

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(1 mark)

(b) Read, from the diagrams in Figure 1 and record in Table 1, the volume of the gas produced for EACH experiment. Experiment 1 has been done for you. **(3 marks)**

(c) (i) Write a **balanced** chemical equation, including state symbols, for the reaction between zinc metal and dilute hydrochloric acid.

Equation: ____

(2 marks)

(ii) Identify the oxidizing agent in the equation in (c) (i), and give a reason, in terms of oxidation numbers, why the agent is oxidizing.

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(2 marks)

(iii) Calculate the maximum volume of hydrogen gas, at room temperature and pressure (RTP), that would be produced when 1.0 gram of zinc metal reacts with excess dilute hydrochloric acid.

(1 mole of a gas occupies 24 dm^3 at RTP, RAM Zn = 65).

____

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(3 marks)

(d) Compare the volume of gas produced in Experiments 2, 3 and 4, to the volume produced in Experiment 1, and give an explanation in EACH case.

Experiment 2: ______________________________

____

Experiment 3: ______________________________

____

Experiment 4: ______________________________

____

(6 marks)

(e) The sketch in Figure 2 shows how the volume of hydrogen varies with time for Experiment 1. In this experiment, an equal number of moles of magnesium granules is used instead of zinc granules.

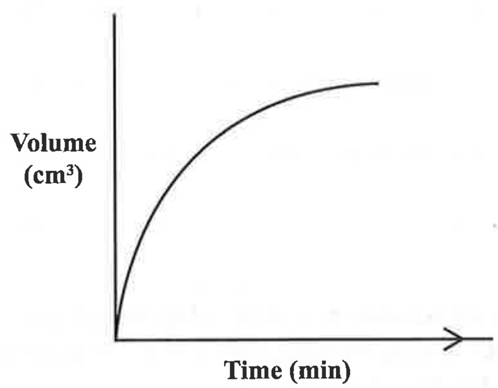

Figure 2. Variation in volume of hydrogen gas produced in Experiment 1 with time

(i) With which granules, zinc or magnesium, will Experiment 1 be more reactive?

____

(1 mark)

(ii) Using the same axes in Figure 2, sketch the curve when magnesium granules are used in Experiment 1.

(1 mark)

(f) Plan and design an experiment that can distinguish between a 'strong electrolyte' and a 'weak electrolyte'. You may select from the range of apparatus and chemicals that is provided below.

Beakers, measuring cylinders, pipettes, burettes, volumetric flasks, stop clocks, bunsen burner, conductivity meter, ammeter, power supply (battery), balance, sodium chloride, barium sulphate, hydrochloric acid, ethanoic (acetic) acid.

Your answer should include the following:

(i) An outline of the steps for a procedure you can use

____

____

____

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____

____

(4 marks)

(ii) A list of the main observations that would be expected at EACH stage of the experiment

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____

____

____

____

(2 marks)

Total 25 marks

2. Both sodium chloride (NaCl) and iodine (I_2) have crystalline structures. A sample of solid NaCl was contaminated with a small amount of iodine crystals. A student suggested that the iodine could be separated from the sodium chloride by gently heating the mixture as shown in Figure 3.

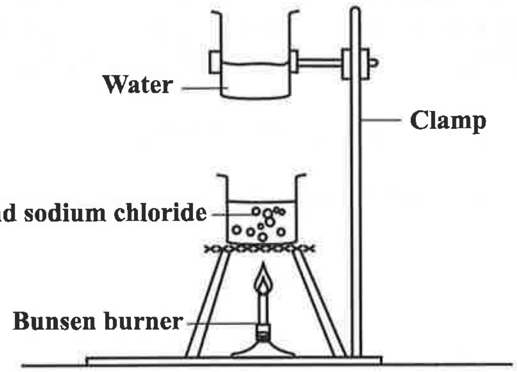


Figure 3. Action of heat on mixture of iodine and sodium chloride

(a) (i) State the name of the process being used in Figure 3 to separate the iodine from sodium chloride.

____

(1 mark)

(ii) Describe what should be observed when the sample of NaCl and I_2 is heated.

____

____

____

(3 marks)

(iii) Using dot cross diagrams, show the bonding in NaCl.

(3 marks)

(iv) By comparing the forces of attraction between the NaCl particles and the I_2 molecules, explain why gentle heating is suitable for separating I_2 from a mixture of NaCl and I_2.

(4 marks)

(b) (i) A second sample of NaCl is known to be contaminated with iron filings (powdered iron). State whether the method outlined in Figure 3 is suitable for separating the Fe from NaCl.

(1 mark)

(ii) With reference to the bonding in Fe, explain your answer to (b) (i).

(3 marks)

Total 15 marks

3. (a) Two compounds, (C_2H_4) and (C_3H_8) labelled **A** and **B** respectively, are gases which burn in oxygen.

(i) Draw the FULLY DISPLAYED structures and state the names of BOTH compounds.

Compound A

Name: ____________________
(2 marks)

Compound B

Name: ____________________
(2 marks)

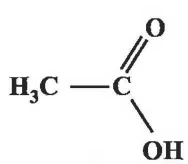
(ii) State whether Compound **A** burns in air with a sooty flame or a clean, blue flame.

____
(1 mark)

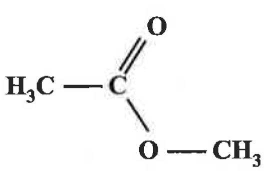
(iii) Write a **balanced** chemical equation for this reaction.

Equation: ____
(2 marks)

(b) The structures of Compounds **C** and **D** are shown below.



Compound C



Compound D

(i) Explain why Compound **C** is soluble in water and Compound **D** is not.

____

____

____

____
(2 marks)

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(ii) Write a **balanced** equation for the chemical reaction of Compound **C** with calcium metal.

Equation: ______________________________

(2 marks)

(c) (i) What is a polymer?

____

____

(1 mark)

(ii) Compound **E**, C_3H_6, undergoes polymerization when heated under pressure.

a) What type of polymerization has taken place?

____

b) What is the name of the product that is formed in (ii) a)?

____

c) State ONE use of the product formed in (ii) a).

____

(3 marks)

Total 15 marks

SECTION B

Answer ALL questions.

4. (a) Explain why magnesium conducts electricity when solid but magnesium iodide (MgI_2) only conducts electricity when molten or in solution. **(4 marks)**

(b) (i) Draw a labelled diagram of apparatus which could be used in the electrolysis of molten MgI_2. **(2 marks)**

(ii) Write equations to indicate the chemical reactions which will occur at EACH electrode. **(4 marks)**

(c) A current of 5 A is passed for 10 minutes through the molten MgI_2. Calculate the mass of product that will be formed at the cathode.

(Relative atomic masses: Mg = 24; I = 127)
(1 Faraday = 96 500 C mol^{-1}) **(5 marks)**

Total 15 marks

Write your answer to Question 4 here.

Write your answer to Question 4 here.

5. (a) Major stores of carbon exist in the atmosphere and the oceans. Carbon is moved through living systems via the carbon cycle. Figure 4 is a diagram of the carbon cycle, with three processes labelled **X**, **Y** and **Z** respectively.

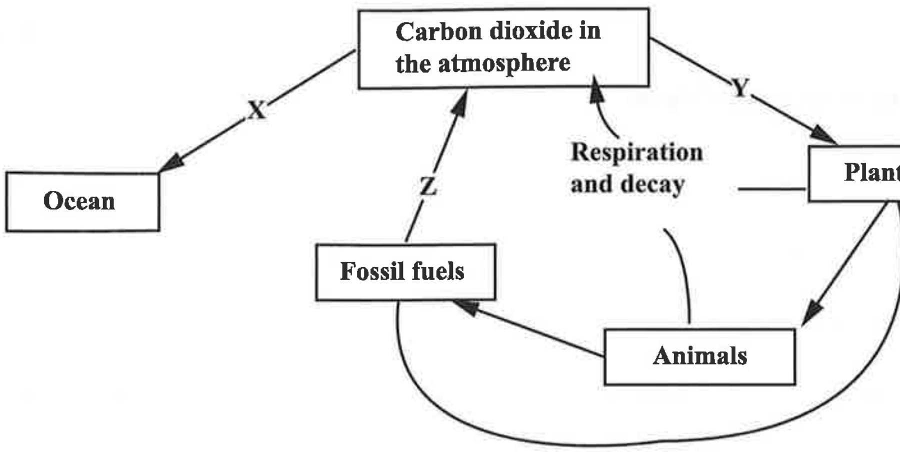

Figure 4. Carbon cycle

(i) Outline how processes **X** and **Y** move carbon through the cycle. **(2 marks)**

(ii) Describe the process occurring at **Z** and explain ONE harmful effect that could occur as a result of this process. Write ONE **balanced** equation to support your answer. **(5 marks)**

(b) A chemist conducted experiments on two unknown metals, **R** and **M**, and found the properties recorded in Table 2.

TABLE 2: PROPERTIES OF R AND M

Properties	**Metal R**	**Metal M**
Formula of oxide	**RO**	**MO**
Reaction with aqueous aluminium ions	No visible reaction	Aluminium is displaced from solution
Reaction with dilute acid	Very slow production of hydrogen gas	Rapid liberation of hydrogen gas

(i) Write a suitable equation for

a) the action of heat on the carbonate of **R**

b) the reaction between **M** and aqueous aluminium ions. **(4 marks)**

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(ii) Explain why **M** is more reactive with dilute acid than **R**. **(2 marks)**

(iii) State ONE physical and ONE chemical property that both **M** and **R** are likely to share. **(2 marks)**

Total 15 marks

Write your answer to Question 5 here.

6. (a) Chlorofluorocarbons and phosphates are two known pollutants.

(i) Identify a possible source for EACH of them. **(2 marks)**

(ii) Explain how the environment is affected by EACH pollutant. **(4 marks)**

(b) The use of landfills, incinerators and recycling are three major methods used for solid waste disposal. Discuss the advantages and disadvantages of using these THREE methods in the Caribbean. For EACH method, identify ONE advantage and TWO disadvantages. **(9 marks)**

Total 15 marks

Write your answer to Question 6 here.